DATA ANALYTICS • CUSTOMER INTELLIGENCE • MACHINE LEARNING

INTELLIGENT CUSTOMER CHURN PREDICTION

Using Data Analytics and Classification Models

PROJECT REVIEW 1
LITERATURE SURVEY | RESEARCH GAPS | PROJECT DIRECTION

UNIVERSITY	Koneru Lakshmaiah Education Foundation (KLH University)
DEPARTMENT	Computer Science and Engineering
COURSE	Data Analytics and Visualization (24DSB3101)
ACADEMIC YEAR	2026–2027

PROJECT TEAM

ALLIPURAM AKHIL 2420030540	**DEVINENI SURYA** 2420030440

FACULTY GUIDE | DR. K. SWAPNIKA

FOUNDATION

01 Project Framing

Refined Project Title

Intelligent Customer Churn Prediction Using Data Analytics and Classification Models

The title captures the project's focus on examining customer data, preparing predictive features, comparing classification algorithms and presenting churn-risk outputs. The literature review spans 2021–2026 and contains 20 selected research papers, as specified in the supplied review presentation.

Problem Statement

Customer churn occurs when a customer discontinues a service or moves to another provider. In subscription-based businesses, churn can affect recurring revenue and increase the effort required to acquire replacement customers. Churn behaviour is not explained by one attribute alone. It can be associated with service usage, tenure, contract type, billing and other customer characteristics. A useful prediction workflow therefore needs more than a single model score: it should examine data quality, class distribution, model performance, risk thresholds and the way predictions are communicated to decision-makers.

This project develops a data analytics and classification workflow for identifying customers who may churn. It compares multiple classifiers under a consistent evaluation approach and aims to connect model outputs with customer-level risk categories and an interactive dashboard. The supplied presentation positions the work around data exploration, preprocessing, classification, evaluation, interpretation and dashboard-based prediction.

Review Method

Review element	How it is applied
Publication window	2021 through 2026, matching the recent five-year literature focus stated in the presentation.
Selection basis	Research papers with publisher/DOI pages or identifiable journal records found during source checking.
Comparison dimensions	Problem/domain, method or research approach, reported contribution/result, and relevance to this project.
Comparability rule	Numerical results belong to each paper's own dataset and validation protocol. They are not treated as directly comparable scores.
Limitation rule	Where a limitation is not explicit in the publisher summary, the review records relevance rather than inventing a limitation.

RELATED WORK • PUBLICATIONS 2021–2026

02.1 Literature Survey

NO. / YEAR	PAPER TITLE & SOURCE	METHOD / RESEARCH APPROACH	KEY CONTRIBUTION & PROJECT RELEVANCE
01 2021	Hybrid approach using machine learning algorithms for customers' churn prediction in the telecommunications industry Beeharry & Fokone | Concurrency and Computation: Practice and Experience	Two-layer flexible voting ensemble; tested on IBM and Duke datasets with balanced and imbalanced settings.	Motivates combining diverse classifiers instead of relying on one model. DOI: 10.1002/cpe.6627
02 2021 online / 2022 issue	Distributed model for customer churn prediction using convolutional neural network Author details not specified in source slide | Journal of Modelling in Management	2-D CNN with preprocessing and Apache Spark distributed processing; Telco Customer Churn data.	Shows a deep-learning and distributed-processing route; reported accuracy is dataset-specific. DOI: 10.1108/JM2-01-2021-0032
03 2022	Detecting the Risk of Customer Churn in Telecom Sector: A Comparative Study Edwine et al. | Mathematical Problems in Engineering	Mutual-information feature selection, controlled-ratio undersampling, Random Forest, SVM, KNN and hyperparameter search.	Addresses class imbalance and model tuning; supports evaluating more than accuracy. DOI: 10.1155/2022/8534739
04 2023	Research on customer churn prediction and model interpretability analysis Peng, Peng & Li | PLOS ONE	Banking churn study combining predictive modelling with interpretability analysis.	Connects churn prediction to explanations for decision-making. DOI: 10.1371/journal.pone.0289724

. / YEAR	PAPER TITLE & SOURCE	METHOD / RESEARCH APPROACH	KEY CONTRIBUTION & PROJECT RELEVANCE
05 2023	Early warning of telecom enterprise customer churn based on ensemble learning Zhou et al. | PLOS ONE	Ensemble-learning-based early-warning approach for enterprise telecom customers.	Highlights early warning as a retention use case rather than a post-churn report. DOI: 10.1371/journal.pone.0292466
06 2023	Customer churning analysis using machine learning algorithms Author details not specified in source slide | International Journal of Intelligent Networks	Compared stochastic gradient boosting, Random Forest, KNN and Logistic Regression.	Useful baseline comparison across linear, neighbour-based and ensemble methods. DOI: 10.1016/j.ijin.2023.05.005
07 2023	Machine learning based customer churn prediction in home appliance rental business Suh | Journal of Big Data	Real operating data from a Korean water-purifier rental company; evaluated F1 and AUC.	Shows churn methods can transfer beyond telecom; reported F1 0.93 and AUC 0.88 in that dataset. DOI: 10.1186/s40537-023-00721-8
08 2024	Customer churn prediction in telecom sector using machine learning techniques Wagh et al. | Results in Control and Optimization	Applied machine-learning techniques to telecom churn data and examined factors associated with leaving.	Supports domain-specific comparison and identifying churn-related behaviour. DOI: 10.1016/j.rico.2023.100342
09 2024	Hybrid black-box classification for customer churn prediction with segmented interpretability analysis Author details not specified in source slide | Decision Support Systems	Hybrid segmented models with black-box classifiers; SHAP-based visualization and benchmark experiments.	Links predictive performance to segment-level interpretation for retention decisions. DOI: 10.1016/j.dss.2024.114217
10 2024	Explaining customer churn prediction in telecom industry using tabular machine learning models Author details not specified in source slide | Machine Learning with Applications	Compared tabular classifiers; used global and local explanations including SHAP and visual plots.	Gradient Boosting reported 81% accuracy; demonstrates explainability alongside performance. DOI: 10.1016/j.mlwa.2024.100567
11 2024	A Review on Machine Learning Methods for Customer Churn Prediction and Recommendations for Business Practitioners Manzoor et al. | IEEE Access	Review of 212 published articles from 2015–2023 across churn workflow stages and industries.	Identifies profit-oriented metrics as under-researched; recommends ensembles and explainability. DOI: 10.1109/ACCESS.2024.3402092
12 2024	A Decade of Churn Prediction Techniques in the TelCo Domain: A Survey Barsotti et al. | SN Computer Science	Survey of telecom churn algorithms, datasets, evaluation metrics, issues and open challenges.	Provides a domain-focused map of methods; notes heterogeneous datasets complicate direct score comparisons. DOI: 10.1007/s42979-024-02722-7
13 2024	Investigating customer churn in banking: a machine learning approach and visualization app for data science and management Author details not specified in source slide | Data Science and Management	Compared machine-learning algorithms and built an R Shiny visualization application for banking churn.	Supports combining model comparison with an interactive dashboard. DOI: 10.1016/j.dsm.2023.09.002
14 2024	Using topological data analysis and machine learning to predict customer churn Sagming, Heymann & Visaya | Journal of Big Data	Explores topological data analysis alongside machine learning for churn prediction.	Introduces structural data analysis as an alternative feature/representation perspective. DOI: 10.1186/s40537-024-01020-6
15 2025	A data-driven approach with explainable artificial intelligence for customer churn prediction in the telecommunications industry Author details not specified in source slide | Results in Engineering	XAI-Churn TriBoost combines XGBoost, CatBoost and LightGBM using soft voting; SHAP and LIME are used.	Reports more than 2 million records and 96.44% accuracy; figures are dataset-specific and not directly comparable. DOI: 10.1016/j.rineng.2025.104629
16 2025	Customer Churn Prediction: A Systematic Review of Recent Advances, Trends, and Challenges in Machine Learning and Deep Learning Imani et al. | Machine Learning and Knowledge Extraction	PRISMA-guided review: 240 studies mapped and 61 examined in detailed synthesis (2020–2024).	Identifies class imbalance, concept drift, explainability and business-aligned metrics as continuing challenges. DOI: 10.3390/make7030105

NO. / YEAR	PAPER TITLE & SOURCE	METHOD / RESEARCH APPROACH	KEY CONTRIBUTION & PROJECT RELEVANCE
17 Online 2025 / issue 2026	A comprehensive survey on customer churn analysis studies Shahabikargara et al. | Journal of Information and Telecommunication	Survey across customer journeys, features, machine-learning algorithms and telecom, banking and gaming sectors.	Useful for framing cross-industry differences and future research directions. DOI: 10.1080/24751839.2025.2528440
18 2025	An adaptive analytics framework for customer retention through integrative feature optimization and ensemble learning Author details not specified in source slide | Decision Analytics Journal	PCA, feature optimization and weighted-voting XGBoost/LightGBM; SHAP interpretation.	Reports AUC 0.99 and accuracy 0.98 on its e-commerce dataset; requires cautious cross-dataset interpretation. DOI: 10.1016/j.dajour.2025.100626
19 2025	Explainable Customer Churn Prediction in Telecom Using Ensemble Learning and SHAP Analysis Rasool, Aljawaheri & Karram | Mathematical Modelling of Engineering Problems	Stacking ensemble of Decision Tree, Random Forest, Extra Trees, Gradient Boosting and XGBoost; SHAP explanations.	Directly relevant to comparing a stacked hybrid with baseline classifiers. DOI: 10.18280/mmep.121226
20 2026	Explainable AI-driven customer churn prediction: a multi-model ensemble approach with SHAP-based feature analysis El Attar & El-Hajj | Frontiers in Artificial Intelligence	Seven models, soft-voting ensemble, SHAP, customer segmentation and threshold optimization on Telco dataset.	Reports XGBoost ROC-AUC 0.932 and ensemble ROC-AUC 0.918; integrates prediction with interpretation and risk profiles. DOI: 10.3389/frai.2026.1748799

SYNTHESIS

03 Literature Synthesis and Research Gaps

What the Recent Literature Shows

- Ensemble methods are widely used in churn prediction, including Random Forest, Gradient Boosting, XGBoost, CatBoost, LightGBM, voting and stacking.
- Deep-learning approaches such as CNN, RNN/LSTM and DNN are explored when sequential, behavioural or larger-scale data are available.
- Explainability is increasingly important. SHAP and LIME can connect a prediction with influential customer attributes.
- Class imbalance and threshold choice affect recall and the number of customers flagged for retention action.
- Dashboard and decision-support studies show the value of connecting model evaluation to usable prediction outputs.
- Reported accuracy and AUC values vary with dataset, preprocessing, class balance and validation design. A fair model comparison needs a shared protocol.

Research Gaps Identified and Project Response

RESEARCH GAP	EVIDENCE FROM LITERATURE	PROJECT RESPONSE
1. End-to-end workflow	Many papers emphasize a model, review or a single stage of the process.	Integrate upload, validation, EDA, preprocessing, modelling, evaluation and a prediction dashboard.
2. Imbalance handling	Class imbalance and resampling recur as concerns in reviews and comparative studies.	Document class distribution and compare suitable class-weighting or resampling strategies inside training folds.
3. Fair model comparison	Different datasets and protocols limit direct comparison across published scores.	Evaluate selected baseline models on a shared split and cross-validation procedure.
4. Actionable risk ranking	A churn label alone does not tell a team which customers to prioritize.	Convert calibrated probabilities into defined risk categories and validate thresholds.
5. Explainability + output	XAI studies show the value of explanations, while operational usability remains important.	Pair feature importance and local explanations with individual and batch prediction outputs.

PROJECT DIRECTION

04 Project Objectives and Proposed Contribution

Project Objectives

- Explore customer churn data and examine missing values, duplicates, class balance and feature distributions.
- Prepare model features through appropriate missing-value treatment, encoding and feature engineering.
- Compare Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, XGBoost and CatBoost using consistent evaluation metrics.
- Interpret influential features and generate churn probabilities with customer-level risk categories.
- Deliver an interactive dashboard for exploratory analysis, model evaluation, single-customer prediction and batch prediction.
- Test a proposed hybrid or ensemble only after its model definition and results are reproducibly validated.

Proposed Contribution

The contribution is the integration of data analysis, consistent classification-model evaluation, churn-risk categorization and dashboard-based prediction in one workflow. The project aims to make the evaluation understandable and operationally useful, not simply to maximize one headline metric.

Scope of the Proposed System

SYSTEM AREA	INTENDED FUNCTION
Data analysis	Dataset overview, quality checks, churn distribution, univariate/bivariate relationships and visual summaries.
Preparation	Missing-value treatment, categorical encoding, feature engineering and a documented imbalance strategy.
Classification	A consistent comparison of the selected baseline algorithms; any hybrid extension is treated as a separate experiment.
Evaluation	Accuracy, precision, recall, F1, ROC-AUC, PR-AUC and confusion-matrix analysis, with threshold choice documented.
Interpretation	Feature importance and local explanation for an individual customer prediction, where the relevant implementation is available.
Dashboard	EDA, model evaluation, individual prediction, batch prediction and prediction history/output.

WORKFLOW

05 Proposed Methodology

The proposed workflow is organized as a sequence of data validation, analysis, preparation, model training, evaluation, interpretation and presentation. The same evaluation protocol should be applied across the selected classifiers so that model differences are attributable to the methods rather than inconsistent data handling.

01	**DATA INPUT** Load the customer churn CSV and validate its schema and target field.
02	**EXPLORATORY ANALYSIS** Examine missing values, churn distribution, feature relationships and visual patterns.
03	**PREPARATION** Apply imputation, encoding, feature engineering and the selected imbalance strategy.
04	**MODEL TRAINING** Train baseline classifiers using training data only.
05	**EVALUATION** Compare accuracy, precision, recall, F1, ROC-AUC, PR-AUC and confusion matrix.
06	**INTERPRETATION** Review feature importance and explain a selected individual customer prediction.
07	**DASHBOARD** Present risk category, individual/batch output and available prediction history.

Methodological Controls

- Fit imputation, encoding, scaling and feature selection on training data only, preferably inside a pipeline or cross-validation fold.
- Keep the test set separate from model selection and threshold tuning.
- Use stratified splitting or cross-validation where appropriate for the churn target.
- Report class prevalence and more than one metric; interpret recall and precision together.
- Clearly distinguish an experiment proposed for future work from one with completed and reproducible outputs.

PROJECT ALIGNMENT

06 Connection to Current Project Results

The uploaded presentation reports the following dataset profile and model findings. These figures are reproduced as stated in that presentation and should be checked against the final execution outputs before final submission.

Dataset Profile Reported in the PPT

DATA CHECK	REPORTED VALUE
Records	5,000 rows
Columns	13 columns
Churners	1,365 customers
Churn prevalence	27.3%
Missing values	150 missing feature values before cleaning
Duplicate rows	0 duplicates reported
Feature transformation	10 input features expanded to 22 encoded features

Reported Metric Leaders

METRIC / SELECTION BASIS	MODEL	VALUE
Highest accuracy	Random Forest	0.744
Highest ROC-AUC	Gradient Boosting	0.7635
Highest PR-AUC	Gradient Boosting	0.5546
Highest recall	Decision Tree	0.6630
Five-fold CV ROC-AUC used for selection	Logistic Regression	0.7706

How to interpret these results

Different classifiers lead on different metrics. Random Forest leads on accuracy, Gradient Boosting leads on ROC-AUC and PR-AUC, Decision Tree leads on recall, and Logistic Regression was selected using five-fold CV ROC-AUC. The report should state why cross-validated ROC-AUC is the chosen selection criterion and how that choice supports the project objective.

Consistency Check Before Final Submission

Model-list mismatch to resolve

The supplied PPT notes that the abstract mentions DNN, 1D-CNN, LSTM and CNN-LSTM, while the reported results list six classical machine-learning models. Confirm which models were actually implemented, then make the abstract, methodology, results tables and conclusions use the same model list.

The project direction slide names Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, XGBoost and CatBoost for comparison. The result-alignment slide provides metric leaders but not a complete metric table for every model. Therefore, this review does not fill in unreported model scores.

EXECUTION PLAN

07 Feasibility, Project Plan and Team

Feasibility Assessment

DIMENSION	ASSESSMENT
Technical feasibility	The project uses tabular customer data and established machine-learning libraries. The workflow can be implemented with Python-based data processing, classification, evaluation and visualization tools.
Economic feasibility	The proposed toolchain relies on open-source software and a publicly available churn dataset. No specialized paid platform is required by the scope described in the presentation.
Operational feasibility	The work is organized into manageable stages: data checks, EDA, preparation, modelling, evaluation, interpretation and dashboard presentation.

Project Plan

PHASE	PLANNED ACTIVITY
Phase 1	Finalize literature review and scope; document dataset source; inspect schema, missing values, duplicates and target distribution.
Phase 2	Complete exploratory analysis, feature preparation, encoding and the selected class-imbalance strategy.
Phase 3	Train selected baseline classifiers and conduct a shared model comparison using consistent validation.
Phase 4	Evaluate risk probabilities and thresholds; test the proposed hybrid only with a reproducible design; analyse errors and influential features.
Phase 5	Verify dashboard modules and explanation outputs; finalize results, documentation, report and presentation.

Team Details

S.NO.	TEAM MEMBER	ROLL NUMBER	PROPOSED RESPONSIBILITY
1	Allipuram Akhil	2420030540	Project coordination, data exploration, preprocessing and model implementation.
2	Devineni Surya	2420030440	Model comparison, evaluation, result interpretation and documentation/presentation.

Faculty Guide

Dr. K. Swapnika | Department of Computer Science and Engineering | KLH University

Review 1 outcome

The literature supports an imbalance-aware, consistently evaluated churn-classification workflow with interpretable outputs. The project's next step is to align the stated model scope with the actually executed experiments and demonstrate the proposed dashboard and explanation components.

REFERENCE INDEX

08 References and DOI Identifiers (1/2)

Bibliographic details and DOI identifiers are reproduced from the uploaded presentation. For entries where author information was not visible in the source slide, that limitation is stated rather than filled by assumption.

[1] Beeharry & Fokone (2021). "Hybrid approach using machine learning algorithms for customers' churn prediction in the telecommunications industry." *Concurrency and Computation: Practice and Experience.* **DOI:** 10.1002/cpe.6627

[2] Author details not specified in source slide (online 2021; issue 2022). "Distributed model for customer churn prediction using convolutional neural network." *Journal of Modelling in Management.* **DOI:** 10.1108/JM2-01-2021-0032

[3] Edwine et al. (2022). "Detecting the Risk of Customer Churn in Telecom Sector: A Comparative Study." *Mathematical Problems in Engineering.* **DOI:** 10.1155/2022/8534739

[4] Peng, Peng & Li (2023). "Research on customer churn prediction and model interpretability analysis." *PLOS ONE.* **DOI:** 10.1371/journal.pone.0289724

[5] Zhou et al. (2023). "Early warning of telecom enterprise customer churn based on ensemble learning." *PLOS ONE.* **DOI:** 10.1371/journal.pone.0292466

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[7] Suh (2023). "Machine learning based customer churn prediction in home appliance rental business." *Journal of Big Data.* **DOI:** 10.1186/s40537-023-00721-8

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[9] Author details not specified in source slide (2024). "Hybrid black-box classification for customer churn prediction with segmented interpretability analysis." *Decision Support Systems.* **DOI:** 10.1016/j.dss.2024.114217

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[11] Manzoor et al. (2024). "A Review on Machine Learning Methods for Customer Churn Prediction and Recommendations for Business Practitioners." *IEEE Access.* **DOI:** 10.1109/ACCESS.2024.3402092

[12] Barsotti et al. (2024). "A Decade of Churn Prediction Techniques in the TelCo Domain: A Survey." *SN Computer Science.* **DOI:** 10.1007/s42979-024-02722-7

[13] Author details not specified in source slide (2024). "Investigating customer churn in banking: a machine learning approach and visualization app for data science and management." *Data Science and Management.* **DOI:** 10.1016/j.dsm.2023.09.002

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[15] Author details not specified in source slide (2025). "A data-driven approach with explainable artificial intelligence for customer churn prediction in the telecommunications industry." *Results in Engineering.* **DOI:** 10.1016/j.rineng.2025.104629

[16] Imani et al. (2025). "Customer Churn Prediction: A Systematic Review of Recent Advances, Trends, and Challenges in Machine Learning and Deep Learning." *Machine Learning and Knowledge Extraction.* **DOI:** 10.3390/make7030105

[17] Shahabikargara et al. (online 2025; issue 2026). "A comprehensive survey on customer churn analysis studies." *Journal of Information and Telecommunication.* **DOI:** 10.1080/24751839.2025.2528440

[18] Author details not specified in source slide (2025). "An adaptive analytics framework for customer retention through integrative feature optimization and ensemble learning." *Decision Analytics Journal.* **DOI:** 10.1016/j.dajour.2025.100626

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END OF PROJECT REVIEW 1